

# Validity Analysis: ARADICE

Target context: Egyptian and Levantine Arabic General-Purpose Assistant Users

Overall risk: **MEDIUM**

## Deployment Context

**Use case:** a general-purpose assistant for answering general questions in Egyptian and Levantine dialects

**Target population:** general Arabic speakers from Egypt and the Levant regions

## Dimension Scores

| Dimension         | Score | Rating           | Confidence |
|-------------------|-------|------------------|------------|
| Input Ontology ✓  | 3     | Moderate gaps    | high       |
| Input Content ■   | 2     | Significant gaps | high       |
| Input Form ■      | 2     | Significant gaps | high       |
| Output Ontology ✓ | 3     | Moderate gaps    | high       |
| Output Content ■  | 2     | Significant gaps | high       |
| Output Form       | 3     | Moderate gaps    | high       |
| Average           | 2.5   |                  |            |

■ = highest concern    ✓ = strongest dimension

## Dimension Key

| Abbr. | Dimension       | Definition                                                                                                   |
|-------|-----------------|--------------------------------------------------------------------------------------------------------------|
| IO    | Input Ontology  | Whether the benchmark's test case categories cover the query types expected in deployment.                   |
| IC    | Input Content   | Whether individual datapoint content is culturally and contextually appropriate for the target region.       |
| IF    | Input Form      | Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.             |
| OO    | Output Ontology | Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries. |
| OC    | Output Content  | Whether ground-truth labels align with the judgments of regional stakeholders.                               |
| OF    | Output Form     | Whether the expected output modality matches regional deployment needs and accessibility.                    |

## Overall Summary

AraDiCE is a strong, well-motivated first attempt at Arabic dialectal benchmarking, and its dialectal scope (Egyptian + Levantine) matches the target deployment exactly. Its cultural benchmark (AraDiCE-Culture) and its preference for ArabicMMLU over MMLU are genuine validity assets for EGY/LEV assistant users. However, for a general-purpose conversational assistant deployment, three structural weaknesses stand out: (1) the core assistant behavior — free-form dialectal generation — was collapsed to a selection task after models failed, so the benchmark does not directly evaluate generation quality; (2) the input/output form is text-only and treats Levantine as monolithic, missing spoken dialect, Arabizi orthography, code-switching, and Syrian/Lebanese/Jordanian/Palestinian sub-varieties; (3) label quality has no IAA statistics, non-uniform QA, authors-as-annotators for parts of the cultural dataset, and documented uneven post-editing effort between Egyptian and Levantine. Additionally, most cognitive benchmarks retain Western-origin content and

TruthfulQA categories, and evaluation is zero-shot with  $\leq 13\text{B}$  parameter models — limiting external validity for production-scale, instruction-tuned assistants.

## Practical Guidance

### What This Benchmark Measures

AraDiCE provides a defensible measurement of (a) whether models can identify and understand Egyptian and Levantine dialectal text, (b) whether models can translate between MSA/English and these dialects, and (c) whether models have region-specific cultural awareness across the nine culturally-marked categories in AraDiCE-Culture (input\_ontology and output\_ontology are its strongest dimensions). The ArabicMMLU-derived world-knowledge component provides culturally grounded evaluation of factual knowledge relevant to Arabic-speaking users.

### Construct Depth

Depth is uneven. Cultural awareness is measured through 30 country-distinct MCQ questions plus qualitative generation analysis — narrow but genuinely region-grounded (strong on AraDiCE-Culture, weak on statistical power). Dialectal generation is measured only via a proxy selection task and qualitative examples [Q57, Q151], leaving the core assistant output behavior largely un-probed. Cognitive tasks (PIQA, OBQA, Winogrande, BoolQ, Belebele, TruthfulQA) reach into commonsense and reading comprehension but retain Western content framings (input\_content and output\_content dimensions weakest). Zero-shot  $\leq 13\text{B}$  evaluation further caps how deeply the results speak to production deployment.

### What Else You Need

For a valid EGY/LEV assistant evaluation, supplement AraDiCE with: (i) free-form dialectal generation evaluation with human rating or a dialect-adequacy-prompted judge, correcting for GPT-4's MSA bias [WEB-16]; (ii) spoken dialect ASR/generation using NADI 2025 or MGB-3 [WEB-15] if voice-enabled; (iii) Levantine sub-variety evaluation via DialectalArabicMMLU [WEB-17]; (iv) code-switching and Arabizi robustness tests (currently unaddressed); (v) larger/proprietary-model comparisons via HELM Arabic [WEB-26]; (vi) regionally-authored misinformation items to replace or augment TruthfulQA categories; (vii) IAA studies on a sample of AraDiCE items to quantify label reliability, especially for Egyptian where more post-editing was required [Q230].

| ID     | Evidence                                                                                                                                                                                                                                                                                                          |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q57    | "To address this, we simplified the task: models were given multiple response options, with only one in the correct dialect, and asked to select the best response." (p.5)                                                                                                                                        |
| Q151   | "We conducted qualitative analysis on the generation task of the models by prompting them in the dialectal version of "How would you respond to this phrase in the X dialect", where X could be one of the Egyptian, Levantine, or Gulf dialects (examples on models' response can be found in Table 12)." (p.19) |
| Q230   | "When comparing post-editing for dialects, we observed that Egyptian Arabic required significantly more edits, which in turn increased the time and effort needed." (p.32)                                                                                                                                        |
| WEB-15 | NADI 2025 addresses spoken output form gap for voice-enabled deployment ( <a href="https://arxiv.org/html/2509.02038">https://arxiv.org/html/2509.02038</a> )                                                                                                                                                     |
| WEB-16 | GPT-4 as evaluator systematically scores MSA higher than dialect, cautioning against naive LLM-as-judge use ( <a href="https://arxiv.org/pdf/2407.07551">https://arxiv.org/pdf/2407.07551</a> )                                                                                                                   |
| WEB-17 | DialectalArabicMMLU separates Syrian dialect from other Levantine, evidencing that a finer output taxonomy for Levantine sub-varieties is possible and reveals gaps AraDiCE's monolithic Levantine label conceals ( <a href="https://arxiv.org/abs/2510.27543">https://arxiv.org/abs/2510.27543</a> )             |
| WEB-26 | Stanford HELM Arabic enables comparison with larger proprietary models beyond AraDiCE's $\leq 13\text{B}$ cap ( <a href="https://crfm.stanford.edu/helm/arabic/latest/">https://crfm.stanford.edu/helm/arabic/latest/</a> )                                                                                       |

# Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: ARADICE | Context: Egyptian and Levantine Arabic General-Purpose Assistant Users

**Input Ontology definition:** Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

AraDiCE covers a broad and generally relevant task taxonomy for a general-purpose assistant deployment: dialect identification, dialectal machine translation, world knowledge, commonsense reasoning, reading comprehension, misinformation, and a novel region-specific cultural awareness benchmark [Q10, Q11, Q14, Q65]. Dialectal evaluation is explicitly scoped to Egyptian and Levantine [Q13], matching the target population. However, key gaps for a general-purpose conversational assistant remain: the dialectal generation task was reformulated from free-form to a selection task after models failed to genuinely produce dialectal responses [Q55, Q56, Q57], which under-represents the core open-ended generation ability an assistant is expected to exercise. TruthfulQA carries misinformation categories (health, law, finance, politics) that were not re-evaluated for regional relevance [Q73, Q121], introducing potential construct-irrelevant variance. The benchmark is entirely text-based [Q16], excluding spoken dialectal input which is the primary daily communication mode of much of the target population [WEB-15].

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q10    | "We primarily evaluate their fundamental NLP abilities in encoding dialects through dialect identification (e.g. El-Haj et al. (2018)) and dialectal machine translation (e.g. Abdul-Mageed et al. (2023)) tasks." (p.2)                                                                                                                                                                                                                 |
| Q11    | "To assess broader cognitive abilities, we include benchmarks that gauge World Knowledge (ArabicMMLU (Koto et al., 2024)), Common Sense Reasoning (PIQA (Bisk et al., 2020), OBQA (Mihaylov et al., 2018), Winogrande (Sakaguchi et al., 2021)), Reading Comprehension, BoolQ (Clark et al., 2019), Belebele (Bandarkar et al., 2024)), and the ability to handle Misinformation (TruthfulQA (Lin et al., 2022))." (p.2)                 |
| Q13    | "We concentrate on the Levantine (LEV), and Egyptian (EGY) dialects." (p.2)                                                                                                                                                                                                                                                                                                                                                              |
| Q14    | "We introduce a third and novel benchmark, AraDiCE-Culture, focused on cultural awareness across the Levantine, Egyptian, and Gulf regions." (p.2)                                                                                                                                                                                                                                                                                       |
| Q16    | "We develop benchmarks through the translation and post-editing of dialectal data, and assess LLMs' performance in cognitive abilities, understanding, and generation tasks across MSA and Dialectal Arabic." (p.2)                                                                                                                                                                                                                      |
| Q55    | "Initially, we used a dialectal response generation task (Naous et al., 2022), where models were prompted in a specific dialect and asked to generate a response." (p.5)                                                                                                                                                                                                                                                                 |
| Q56    | "However, qualitative analysis revealed that models often explained the prompt instead of generating a dialectal response (see Appendix C.2 for examples)." (p.5)                                                                                                                                                                                                                                                                        |
| Q57    | "To address this, we simplified the task: models were given multiple response options, with only one in the correct dialect, and asked to select the best response." (p.5)                                                                                                                                                                                                                                                               |
| Q65    | "We now assess the dialectal Arabic abilities in LLMs through world knowledge, reading comprehension, commonsense reasoning, and misinformation tasks, providing a thorough evaluation of their performance on dialect-specific challenges." (p.6)                                                                                                                                                                                       |
| Q73    | "The results for the TruthfulQA task across Lev, Egy, and MSA show that model performances are often close to random, especially for Lev and Egy, as reported in Figure 7." (p.7)                                                                                                                                                                                                                                                        |
| Q121   | "For assessing misinformation, we used TruthfulQA (Lin et al., 2022). This dataset is specifically designed to evaluate the accuracy and truthfulness of answers provided by language models. It includes 817 questions across 28 diverse categories, such as health, law, finance, and politics. The questions are crafted to address the potential for incorrect answers stemming from common misconceptions or false beliefs." (p.17) |
| WEB-15 | NADI 2025 addresses spoken output form gap for voice-enabled deployment ( <a href="https://arxiv.org/html/2509.02038">https://arxiv.org/html/2509.02038</a> )                                                                                                                                                                                                                                                                            |

## Strengths

- Task taxonomy spans both linguistic encoding (dialect ID, MT) and higher-order cognitive abilities (world knowledge, commonsense, RC), matching the range of a general-purpose assistant [Q10, Q11, Q65]
- Dialectal scope explicitly restricted to Egyptian and Levantine, directly matching deployment target [Q13]
- Novel cultural awareness benchmark adds an evaluation dimension particularly relevant to Egyptian/Levantine users [Q14, Q17, Q122]

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q10  | "We primarily evaluate their fundamental NLP abilities in encoding dialects through dialect identification (e.g. El-Haj et al. (2018)) and dialectal machine translation (e.g. Abdul-Mageed et al. (2023)) tasks." (p.2)                                                                                                                                                                                                                                                                                                                                                                               |
| Q11  | "To assess broader cognitive abilities, we include benchmarks that gauge World Knowledge (ArabicMMLU (Koto et al., 2024)), Common Sense Reasoning (PIQA (Bisk et al., 2020), OBQA (Mihaylov et al., 2018), Winogrande (Sakaguchi et al., 2021)), Reading Comprehension, BoolQ (Clark et al., 2019), Belebele (Bandarkar et al., 2024)), and the ability to handle Misinformation (TruthfulQA (Lin et al., 2022))." (p.2)                                                                                                                                                                               |
| Q13  | "We concentrate on the Levantine (LEV), and Egyptian (EGY) dialects." (p.2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Q14  | "We introduce a third and novel benchmark, AraDiCE-Culture, focused on cultural awareness across the Levantine, Egyptian, and Gulf regions." (p.2)                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Q17  | "We create the first benchmark for region-wise cultural evaluation in the Gulf, Egyptian, and Levant regions." (p.2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Q65  | "We now assess the dialectal Arabic abilities in LLMs through world knowledge, reading comprehension, commonsense reasoning, and misinformation tasks, providing a thorough evaluation of their performance on dialect-specific challenges." (p.6)                                                                                                                                                                                                                                                                                                                                                     |
| Q122 | "In this paper, we introduce a novel benchmark focused on cultural awareness across the Levant, Egypt, and Gulf regions. This benchmark goes beyond language proficiency to assess whether LLMs can accurately capture the cultural nuances specific to these regions. The dataset contains 180 questions covering 9 cultural aspects, including events (e.g., public holidays), traditions (e.g., food and clothing), geography, history, literature, and religion. To our knowledge, this is the first cultural dataset specifically designed to capture region-wise cultural understanding." (p.17) |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                            |
|------|----------------------------------------------------------------|
| IO-1 | Identify test case categories required for regional deployment |
| IO-2 | Check if taxonomy omits regionally relevant categories         |
| IO-3 | Check if taxonomy includes irrelevant categories               |
| IO-4 | Document category gaps harming content validity                |

### Checklist responses:

**IO-1:** Required categories for a general-purpose EGY/LEV assistant include: free-form dialectal generation, dialectal comprehension/QA, cultural/regional knowledge, commonsense reasoning, and reading comprehension. AraDiCE covers dialect ID, MT, cognitive tasks, and cultural awareness [Q10, Q11, Q14, Q65]; spoken dialect is absent [Q16], and free-form generation was collapsed to selection [Q57].

**IO-2:** Yes — free-form dialectal generation is under-represented (converted to selection) [Q55, Q56, Q57]; spoken dialectal input/output is absent [Q16]; code-switching (dialect↔MSA↔English/French) is not explicitly evaluated as a task category (INSUFFICIENT DOCUMENTATION on code-switching evaluation).

**IO-3:** TruthfulQA misinformation categories (health, law, finance, politics) were not re-evaluated for regional relevance [Q121], and models score near-random on dialectal TruthfulQA [Q73], suggesting possible construct-irrelevant variance for this deployment. Gulf-region components of AraDiCE-Culture are outside the deployment scope [Q4, Q14] but do not directly harm EGY/LEV validity.

**IO-4:** Category gaps: (a) open-ended dialectal generation [Q57]; (b) spoken dialect [Q16, WEB-15]; (c) code-switching handling; (d) unvetted misinformation category relevance [Q121]. These harm content validity for the assistant use case.

| ID | Evidence |
|----|----------|
|----|----------|

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q4     | "Additionally, we introduce the first-ever fine-grained benchmark designed to evaluate cultural awareness across the Gulf, Egypt, and Levant regions, providing a novel dimension to LLM evaluation." (p.1)                                                                                                                                                                                                                              |
| Q10    | "We primarily evaluate their fundamental NLP abilities in encoding dialects through dialect identification (e.g. El-Haj et al. (2018)) and dialectal machine translation (e.g. Abdul-Mageed et al. (2023)) tasks." (p.2)                                                                                                                                                                                                                 |
| Q11    | "To assess broader cognitive abilities, we include benchmarks that gauge World Knowledge (ArabicMMLU (Koto et al., 2024)), Common Sense Reasoning (PIQA (Bisk et al., 2020), OBQA (Mihaylov et al., 2018), Winogrande (Sakaguchi et al., 2021)), Reading Comprehension, BoolQ (Clark et al., 2019), Belebele (Bandarkar et al., 2024)), and the ability to handle Misinformation (TruthfulQA (Lin et al., 2022))." (p.2)                 |
| Q13    | "We concentrate on the Levantine (LEV), and Egyptian (EGY) dialects." (p.2)                                                                                                                                                                                                                                                                                                                                                              |
| Q14    | "We introduce a third and novel benchmark, AraDiCE-Culture, focused on cultural awareness across the Levantine, Egyptian, and Gulf regions." (p.2)                                                                                                                                                                                                                                                                                       |
| Q16    | "We develop benchmarks through the translation and post-editing of dialectal data, and assess LLMs' performance in cognitive abilities, understanding, and generation tasks across MSA and Dialectal Arabic." (p.2)                                                                                                                                                                                                                      |
| Q55    | "Initially, we used a dialectal response generation task (Naous et al., 2022), where models were prompted in a specific dialect and asked to generate a response." (p.5)                                                                                                                                                                                                                                                                 |
| Q56    | "However, qualitative analysis revealed that models often explained the prompt instead of generating a dialectal response (see Appendix C.2 for examples)." (p.5)                                                                                                                                                                                                                                                                        |
| Q57    | "To address this, we simplified the task: models were given multiple response options, with only one in the correct dialect, and asked to select the best response." (p.5)                                                                                                                                                                                                                                                               |
| Q65    | "We now assess the dialectal Arabic abilities in LLMs through world knowledge, reading comprehension, commonsense reasoning, and misinformation tasks, providing a thorough evaluation of their performance on dialect-specific challenges." (p.6)                                                                                                                                                                                       |
| Q73    | "The results for the TruthfulQA task across Lev, Egy, and MSA show that model performances are often close to random, especially for Lev and Egy, as reported in Figure 7." (p.7)                                                                                                                                                                                                                                                        |
| Q121   | "For assessing misinformation, we used TruthfulQA (Lin et al., 2022). This dataset is specifically designed to evaluate the accuracy and truthfulness of answers provided by language models. It includes 817 questions across 28 diverse categories, such as health, law, finance, and politics. The questions are crafted to address the potential for incorrect answers stemming from common misconceptions or false beliefs." (p.17) |
| WEB-15 | NADI 2025 addresses spoken output form gap for voice-enabled deployment ( <a href="https://arxiv.org/html/2509.02038">https://arxiv.org/html/2509.02038</a> )                                                                                                                                                                                                                                                                            |

## Information Gaps

- Whether AraDiCE evaluates code-switched input (dialect↔MSA↔English/French) as a distinct task category is not documented
- Whether TruthfulQA category coverage matches misinformation salient to EGY/LEV users (e.g., political, health, conflict-related regional misinformation)

## Requires Expert Verification

- Whether the collapse from free-form generation to selection is an acceptable proxy for assistant generation ability

## Remediation

**Gap 1: Free-form dialectal generation collapsed to selection task; spoken dialect and code-switching absent [Q57, Q16]**

**Recommendation:** Add a supplementary free-form dialectal generation benchmark evaluated with dialect-adequacy-prompted human or LLM judges (accounting for known GPT-4 pro-MSA bias [WEB-16]); pair with NADI 2025 spoken-dialect evaluation [WEB-15] for voice-enabled deployments.

| ID  | Evidence                                                                                                                                                                                                            |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q16 | "We develop benchmarks through the translation and post-editing of dialectal data, and assess LLMs' performance in cognitive abilities, understanding, and generation tasks across MSA and Dialectal Arabic." (p.2) |
| Q57 | "To address this, we simplified the task: models were given multiple response options, with only one in the correct dialect, and asked to select the best response." (p.5)                                          |

|               |                                                                                                                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WEB-15</b> | NADI 2025 addresses spoken output form gap for voice-enabled deployment ( <a href="https://arxiv.org/html/2509.02038">https://arxiv.org/html/2509.02038</a> )                                   |
| <b>WEB-16</b> | GPT-4 as evaluator systematically scores MSA higher than dialect, cautioning against naive LLM-as-judge use ( <a href="https://arxiv.org/pdf/2407.07551">https://arxiv.org/pdf/2407.07551</a> ) |

# Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: ARADICE | Context: Egyptian and Levantine Arabic General-Purpose Assistant Users

**Input Content definition:** Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The benchmark documents an explicit and important limitation: 'most datasets, except ArabicMMLU, are adapted from English and thus are influenced by Western culture' [Q99], and Western cultural design of source content was 'consistently observed' during post-editing [Q224]. Despite post-editing efforts to align translations with Arabic nuance [Q41] and specific adjustments to culturally sensitive Winogrande instances [Q43, Q223], the underlying content largely originates outside the EGY/LEV cultural context. ArabicMMLU is a genuine strength — it was preferred over MMLU precisely because it better represents Arabic-speaking communities [Q42, Q112]. AraDiCE-Culture was constructed from scratch by native annotators from the target regions with search-verified region-distinct answers [Q24, Q25], providing high-fit content for the deployment. However, BoolQ was filtered to passages of 41–76 characters to manage post-editing workload [Q221, Q222], which is a small and unnaturalistic length for reading comprehension. Some TruthfulQA items language-locked to English were dropped [Q225, Q226], reducing coverage. Combined, these produce mixed content validity: strong for ArabicMMLU and AraDiCE-Culture, weaker for the other translated tasks.

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                     |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q24  | "Cultural Dataset: We curated a dataset of 180 culturally specific questions by hiring native annotators from the Gulf, Egypt, and the Levant regions to generate seed questions centered on cultural and country-specific themes." (p.3)                                                                                                                                                    |
| Q25  | "After reviewing all submitted questions, we selected those with varying answers across regions. To verify these differences, we appended a country name to each question and used Google Search to examine the top 5 search results. Only questions with distinct answers across countries were retained, resulting in a final set of 30 unique questions." (p.3)                           |
| Q41  | "To address this, our approach emphasizes refining translations to ensure alignment with Arabic linguistic nuances and cultural relevance." (p.4)                                                                                                                                                                                                                                            |
| Q42  | "For tasks requiring world knowledge, we opted for ArabicMMLU (originally in Modern Standard Arabic, or MSA) instead of MMLU, as it better represents the knowledge and contexts relevant to Arabic-speaking communities and regions." (p.4)                                                                                                                                                 |
| Q43  | "For the Winogrande dataset, we made specific adjustments to culturally sensitive instances, as detailed in Section H.1.3." (p.4)                                                                                                                                                                                                                                                            |
| Q99  | "Another issue to highlight is that most datasets, except ArabicMMLU, are adapted from English and thus are influenced by Western culture. While we made efforts in annotation guidelines and post-editing to address these cultural biases, the subjective nature of sensitivity means that some samples may still be considered sensitive by different individuals or communities." (p.10) |
| Q112 | "It consists of subjects featuring questions written in MSA, sourced from eight different countries, and covers diverse disciplines such as history, geography, law, civics education, and driving tests across different education levels." (p.16)                                                                                                                                          |
| Q221 | "In an attempt to reduce the effort and the time needed to finish post editing all the dataset, we opted to select only 1K samples from BoolQ. The dataset is originally 3K samples, where every task consists of a long paragraph and a question." (p.32)                                                                                                                                   |
| Q222 | "We opted to choose samples where the length of the passage was 41-76 characters." (p.32)                                                                                                                                                                                                                                                                                                    |
| Q223 | "During our analysis of the datasets, we noticed that some samples in Winogrande conflicted with the religion and culture of the Arab region. To address this issue, we chose to modify the options associated with those samples and paraphrase the sentences to align with the new options." (p.32)                                                                                        |
| Q224 | "In all the datasets that were post-edited, except for ArabicMMLU, we consistently observed that the samples were designed with a focus on Western culture." (p.32)                                                                                                                                                                                                                          |
| Q225 | "Another major challenge was handling samples that were language-specific. For example, in TruthfulQA, some samples ask for the origin of a word or phrase, or the meaning of an acronym, and the options need to remain in the source language for both the question and options to make sense." (p.32)                                                                                     |

|      |                                                                                                                                                                                                                                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q226 | "In these cases, code-switching was not a viable option because the phrases were long and would require transliteration, which would make them nonsensical. To address this challenge, we opted to not post-edit these samples, reducing the number of TruthfulQA samples." (p.32) |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Strengths

- Deliberate preference for ArabicMMLU over MMLU on cultural-relevance grounds [Q42]
- AraDiCE-Culture constructed by native EGY/LEV/Gulf annotators with search-verified region-distinct answers [Q24, Q25, Q27]
- Explicit acknowledgement of Western content bias and attempts at mitigation via guidelines and Winogrande edits [Q41, Q43, Q99, Q223]
- Dialectal restriction to EGY/LEV matches deployment target [Q13]

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                     |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q13  | "We concentrate on the Levantine (LEV), and Egyptian (EGY) dialects." (p.2)                                                                                                                                                                                                                                                                                                                  |
| Q24  | "Cultural Dataset: We curated a dataset of 180 culturally specific questions by hiring native annotators from the Gulf, Egypt, and the Levant regions to generate seed questions centered on cultural and country-specific themes." (p.3)                                                                                                                                                    |
| Q25  | "After reviewing all submitted questions, we selected those with varying answers across regions. To verify these differences, we appended a country name to each question and used Google Search to examine the top 5 search results. Only questions with distinct answers across countries were retained, resulting in a final set of 30 unique questions." (p.3)                           |
| Q27  | "For gold-standard answers, we followed the NativeQA framework (Hasan et al., 2024). Annotators reviewed both the questions and the top 5 search results, combining web data with their cultural knowledge to provide the most accurate answers." (p.3)                                                                                                                                      |
| Q41  | "To address this, our approach emphasizes refining translations to ensure alignment with Arabic linguistic nuances and cultural relevance." (p.4)                                                                                                                                                                                                                                            |
| Q42  | "For tasks requiring world knowledge, we opted for ArabicMMLU (originally in Modern Standard Arabic, or MSA) instead of MMLU, as it better represents the knowledge and contexts relevant to Arabic-speaking communities and regions." (p.4)                                                                                                                                                 |
| Q43  | "For the Winogrande dataset, we made specific adjustments to culturally sensitive instances, as detailed in Section H.1.3." (p.4)                                                                                                                                                                                                                                                            |
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## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                                        |
|------|----------------------------------------------------------------------------|
| IC-1 | Do inputs require region-specific cultural/geographic/dialectal knowledge? |
| IC-2 | Does culturally sensitive content align with target culture?               |
| IC-3 | Flag inputs assuming Western-specific knowledge                            |
| IC-4 | Regional annotator recruitment for culturally sensitive instances          |
| IC-5 | Document content issues harming content validity                           |

### Checklist responses:

**IC-1:** Yes — culturally specific queries in AraDiCE-Culture explicitly require EGY/LEV knowledge [Q24, Q25]; ArabicMMLU includes country-sourced content including EGY/LEV [Q112]. Other translated benchmarks (PIQA, OBQA, Winogrande, BoolQ, Belebele, TruthfulQA) do not natively require regional knowledge.

**IC-2:** Partial. AraDiCE-Culture aligns with target culture by design [Q24, Q25]. Winogrande received targeted cultural edits [Q43, Q223]. However, authors themselves acknowledge Western cultural bias persists in most datasets despite

post-editing [Q99, Q224].

**IC-3:** Yes — most non-ArabicMMLU datasets encode Western-culture assumptions [Q99, Q224]; TruthfulQA has language-locked English items that had to be dropped [Q225, Q226]; BoolQ was filtered to short passages (41–76 chars) [Q221, Q222] which may not reflect naturalistic reading tasks.

**IC-4:** Native EGY/LEV/Gulf annotators were recruited for AraDiCE-Culture [Q24] and 32 native-speaker post-editors worked on translated datasets [Q34, Q35]. Winogrande cultural sensitivity issues were surfaced during annotation [Q223]. This partially satisfies the checklist item.

**IC-5:** Content issues harming validity: (a) persistent Western bias in translated datasets [Q99, Q224]; (b) TruthfulQA content dropped for language-lock reasons, reducing coverage [Q225, Q226]; (c) BoolQ length filtering may bias reading comprehension estimates [Q221, Q222]; (d) Egyptian required significantly more post-editing than Levantine [Q230], suggesting uneven content quality across the two target dialects.

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                                     |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q24    | “Cultural Dataset: We curated a dataset of 180 culturally specific questions by hiring native annotators from the Gulf, Egypt, and the Levant regions to generate seed questions centered on cultural and country-specific themes.” (p.3)                                                                                                                                                    |
| Q25    | “After reviewing all submitted questions, we selected those with varying answers across regions. To verify these differences, we appended a country name to each question and used Google Search to examine the top 5 search results. Only questions with distinct answers across countries were retained, resulting in a final set of 30 unique questions.” (p.3)                           |
| Q34    | “Team and Setup The translation team comprised 32 native speakers fluent in Levantine, Egyptian, and MSA, with educational backgrounds ranging from Bachelor’s to Master’s degrees, and ages between 21 and 53.” (p.4)                                                                                                                                                                       |
| Q35    | “Many were professional translators. They received specific guidelines and training tailored to each dataset and dialect to ensure translation quality.” (p.4)                                                                                                                                                                                                                               |
| Q43    | “For the Winogrande dataset, we made specific adjustments to culturally sensitive instances, as detailed in Section H.1.3.” (p.4)                                                                                                                                                                                                                                                            |
| Q99    | “Another issue to highlight is that most datasets, except ArabicMMLU, are adapted from English and thus are influenced by Western culture. While we made efforts in annotation guidelines and post-editing to address these cultural biases, the subjective nature of sensitivity means that some samples may still be considered sensitive by different individuals or communities.” (p.10) |
| Q112   | “It consists of subjects featuring questions written in MSA, sourced from eight different countries, and covers diverse disciplines such as history, geography, law, civics education, and driving tests across different education levels.” (p.16)                                                                                                                                          |
| Q221   | “In an attempt to reduce the effort and the time needed to finish post editing all the dataset, we opted to select only 1K samples from BoolQ. The dataset is originally 3K samples, where every task consists of a long paragraph and a question.” (p.32)                                                                                                                                   |
| Q222   | “We opted to choose samples where the length of the passage was 41-76 characters.” (p.32)                                                                                                                                                                                                                                                                                                    |
| Q223   | “During our analysis of the datasets, we noticed that some samples in Winogrande conflicted with the religion and culture of the Arab region. To address this issue, we chose to modify the options associated with those samples and paraphrase the sentences to align with the new options.” (p.32)                                                                                        |
| Q224   | “In all the datasets that were post-edited, except for ArabicMMLU, we consistently observed that the samples were designed with a focus on Western culture.” (p.32)                                                                                                                                                                                                                          |
| Q225   | “Another major challenge was handling samples that were language-specific. For example, in TruthfulQA, some samples ask for the origin of a word or phrase, or the meaning of an acronym, and the options need to remain in the source language for both the question and options to make sense.” (p.32)                                                                                     |
| Q226   | “In these cases, code-switching was not a viable option because the phrases were long and would require transliteration, which would make them nonsensical. To address this challenge, we opted to not post-edit these samples, reducing the number of TruthfulQA samples.” (p.32)                                                                                                           |
| Q230   | “When comparing post-editing for dialects, we observed that Egyptian Arabic required significantly more edits, which in turn increased the time and effort needed.” (p.32)                                                                                                                                                                                                                   |
| WEB-17 | DialectalArabicMMLU separates Syrian dialect from other Levantine, evidencing that a finer output taxonomy for Levantine sub-varieties is possible and reveals gaps AraDiCE’s monolithic Levantine label conceals ( <a href="https://arxiv.org/abs/2510.27543">https://arxiv.org/abs/2510.27543</a> )                                                                                        |

## Information Gaps

- The distribution of content topics within the post-edited datasets is not broken down by cultural relevance to EGY/LEV specifically
- Whether Levantine content preserves sub-variety references (Syrian vs. Lebanese vs. Jordanian/Palestinian) is not documented

## Requires Expert Verification

- Whether AraDiCE-Culture's 30 seed questions × 6 countries adequately samples the cultural surface area a general-purpose assistant would encounter
  - Whether Winogrande cultural adjustments are sufficient to make it valid for Arab-region use
- 

## Remediation

### Gap 1: Western cultural framing persists in most translated datasets despite post-editing [Q99, Q224]

**Recommendation:** Systematically audit PIQA/OBQA/Winogrande/BoolQ/TruthfulQA items with EGY/LEV native reviewers; expand AraDiCE-Culture beyond 30 seed questions × 6 countries; incorporate regionally-authored items (e.g., PALM 2025 [WEB-27]).

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                                     |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q99    | "Another issue to highlight is that most datasets, except ArabicMMLU, are adapted from English and thus are influenced by Western culture. While we made efforts in annotation guidelines and post-editing to address these cultural biases, the subjective nature of sensitivity means that some samples may still be considered sensitive by different individuals or communities." (p.10) |
| Q224   | "In all the datasets that were post-edited, except for ArabicMMLU, we consistently observed that the samples were designed with a focus on Western culture." (p.32)                                                                                                                                                                                                                          |
| WEB-27 | <a href="https://arxiv.org/pdf/2510.27543">https://arxiv.org/pdf/2510.27543</a>                                                                                                                                                                                                                                                                                                              |

# Input Form (IF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: ARADICE | Context: Egyptian and Levantine Arabic General-Purpose Assistant Users

**Input Form definition:** Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

AraDiCE is entirely text-based [Q16], using written dialectal Arabic in Arabic script alongside MSA and English prompts [Q48]. For a text-based conversational assistant this is broadly appropriate, but the deployment population's daily dialectal communication is predominantly spoken [Q28], and the population also writes dialect in unstandardized orthography including Arabizi/Latin-script transliteration and code-switches with English/French — none of which are documented as being represented in the input form. The benchmark's post-editing guidelines specify careful handling of Quranic verses, hadith, scientific terms, and blanks [Q179, Q184, Q192], which is appropriate for reference-corpus construction but does not address orthographic variation users produce in practice. Passage length filtering in BoolQ [Q222] further constrains input form to short passages. NADI 2025 and MGB-3 exist as complements for spoken dialect [WEB-15] but AraDiCE itself does not address that form.

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q16    | "We develop benchmarks through the translation and post-editing of dialectal data, and assess LLMs' performance in cognitive abilities, understanding, and generation tasks across MSA and Dialectal Arabic." (p.2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Q28    | "Machine Translation is crucial for developing resources for low-resource languages with multiple dialects, like Arabic, which has MSA for formal use and diverse dialects like Egyptian and Levantine for everyday communication (Durrani et al., 2014). These dialects differ significantly from MSA and from each other, presenting unique challenges for NLP tasks." (p.3)                                                                                                                                                                                                                                                                                                                                                                                          |
| Q48    | "We used prompts in English, MSA, and dialects depending on the task, and released both the prompts and configurations via lm-harness." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Q179   | "Not all parts of the questions and the options should be expressed in dialect. Changing those parts or expressing them in dialect will affect the meaning of the question. These parts include (i) Verses from Quran, or lines from Hadith, (ii) Words or the phrases that the question is asking the meaning of, (iii) Words/ phrases that refer to scientific domains, (iv) Technical words or phrases, (v) Lines from poems, (vi) Equations (chemistry or mathematical). While editing, avoid changing these parts, and only edit the other parts that should be expressed in dialect and will not change the semantic meaning of the question. However, edits can be made to address any missing parts or grammatical errors that the question might have." (p.23) |
| Q184   | "Verses from Qur'an should not be modified. Any changes to such elements will alter the meaning of the question. However, in this case, the options can be expressed differently in the dialect." (p.25)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Q192   | "Therefore, it was decided that the context should remain in MSA, as changing it would affect the meaning of the question." (p.28)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Q222   | "We opted to choose samples where the length of the passage was 41-76 characters." (p.32)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| WEB-15 | NADI 2025 addresses spoken output form gap for voice-enabled deployment ( <a href="https://arxiv.org/html/2509.02038">https://arxiv.org/html/2509.02038</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Strengths

- Arabic script consistently used across MSA, EGY, LEV inputs [Q16, Q48]
- Detailed guidelines for handling Arabic-specific text elements (Quranic verses, hadith, scientific terms) [Q179, Q184]
- Explicit acknowledgement of MSA↔dialect distinction and post-editing rules that respect dialectal spelling differences [Q219]

| ID  | Evidence                                                                                                                                                                                                            |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q16 | "We develop benchmarks through the translation and post-editing of dialectal data, and assess LLMs' performance in cognitive abilities, understanding, and generation tasks across MSA and Dialectal Arabic." (p.2) |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q48</b>  | "We used prompts in English, MSA, and dialects depending on the task, and released both the prompts and configurations via lm-harness." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Q179</b> | "Not all parts of the questions and the options should be expressed in dialect. Changing those parts or expressing them in dialect will affect the meaning of the question. These parts include (i) Verses from Quran, or lines from Hadith, (ii) Words or the phrases that the question is asking the meaning of, (iii) Words/ phrases that refer to scientific domains, (iv) Technical words or phrases, (v) Lines from poems, (vi) Equations (chemistry or mathematical). While editing, avoid changing these parts, and only edit the other parts that should be expressed in dialect and will not change the semantic meaning of the question. However, edits can be made to address any missing parts or grammatical errors that the question might have." (p.23) |
| <b>Q184</b> | "Verses from Qur'an should not be modified. Any changes to such elements will alter the meaning of the question. However, in this case, the options can be expressed differently in the dialect." (p.25)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Q219</b> | "For MSA, we prioritized correcting grammatical errors. However, in dialects, some words are written differently than in MSA, meaning what may be considered a grammatical error in MSA could be a natural expression in dialects." (p.32)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                          |
|-------------|--------------------------------------------------------------|
| <b>IF-1</b> | Compare signal distributions between source and target       |
| <b>IF-2</b> | Check if regional infrastructure supports data capture specs |
| <b>IF-3</b> | Identify domain-specific form differences                    |
| <b>IF-4</b> | Document form mismatches harming external validity           |

### Checklist responses:

**IF-1:** Signal form is text-only [Q16]; deployment population uses spoken dialect heavily [Q28] and produces unstandardized written dialect including Arabizi. AraDiCE inputs use standardized Arabic-script post-edited text, likely cleaner than real user input. INSUFFICIENT DOCUMENTATION on Arabizi or orthographic variation representation.

**IF-2:** Text-only inputs are compatible with the mobile-first, low-bandwidth infrastructure of the target region [WEB-9]. However, if the deployment is voice-enabled, AraDiCE cannot evaluate that path.

**IF-3:** Domain-specific form differences: BoolQ passage lengths filtered to 41–76 characters [Q222] — much shorter than naturalistic reading tasks. ArabicMMLU maintains original MSA context for reading-comprehension questions [Q191, Q192]. Code-switching form (dialect+English/French) is not documented as evaluated.

**IF-4:** Form mismatches harming external validity: (a) no spoken input coverage [Q16, Q28, WEB-15]; (b) no Arabizi/Latin-script dialectal orthography evaluation (INSUFFICIENT DOCUMENTATION); (c) no explicit robustness to unstandardized dialectal spelling; (d) BoolQ short-passage constraint [Q222].

| ID            | Evidence                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q16</b>    | "We develop benchmarks through the translation and post-editing of dialectal data, and assess LLMs' performance in cognitive abilities, understanding, and generation tasks across MSA and Dialectal Arabic." (p.2)                                                                                                                                                            |
| <b>Q28</b>    | "Machine Translation is crucial for developing resources for low-resource languages with multiple dialects, like Arabic, which has MSA for formal use and diverse dialects like Egyptian and Levantine for everyday communication (Durrani et al., 2014). These dialects differ significantly from MSA and from each other, presenting unique challenges for NLP tasks." (p.3) |
| <b>Q48</b>    | "We used prompts in English, MSA, and dialects depending on the task, and released both the prompts and configurations via lm-harness." (p.5)                                                                                                                                                                                                                                  |
| <b>Q191</b>   | "For this discipline in ArabicMMLU, the question requires information to be extracted from the 'context,' similar to a standard reading comprehension task." (p.28)                                                                                                                                                                                                            |
| <b>Q192</b>   | "Therefore, it was decided that the context should remain in MSA, as changing it would affect the meaning of the question." (p.28)                                                                                                                                                                                                                                             |
| <b>Q222</b>   | "We opted to choose samples where the length of the passage was 41-76 characters." (p.32)                                                                                                                                                                                                                                                                                      |
| <b>WEB-9</b>  | Egypt reached 97 mobile subscriptions per 100 inhabitants, indicating mobile-first (and often voice-first) usage patterns ( <a href="https://www.statista.com/statistics/502043/mobile-cellular-subscriptions-per-100-inhabitants-in-egypt/">https://www.statista.com/statistics/502043/mobile-cellular-subscriptions-per-100-inhabitants-in-egypt/</a> )                      |
| <b>WEB-15</b> | NADI 2025 addresses spoken output form gap for voice-enabled deployment ( <a href="https://arxiv.org/html/2509.02038">https://arxiv.org/html/2509.02038</a> )                                                                                                                                                                                                                  |

## Information Gaps

- Whether AraDiCE evaluates model robustness to unstandardized dialectal spelling or Arabizi transliteration
- Whether inputs preserve intra-Levantine sub-variety orthographic markers

## Requires Expert Verification

- Whether the character-length filter on BoolQ passages materially distorts reading-comprehension validity
- 

## Remediation

**Gap 1: Text-only inputs, standardized Arabic script, no evaluation of Arabizi / unstandardized dialectal orthography or code-switching**

**Recommendation:** Add held-out sets in Arabizi and code-switched EGY/LEV+English/French to measure model robustness to real user-input distributions; document any evaluation of intra-Levantine sub-variety orthography.

# Output Ontology (OO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: ARADICE | Context: Egyptian and Levantine Arabic General-Purpose Assistant Users

**Output Ontology definition:** Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Output taxonomies mix genuine regional grounding and Western-origin structures. AraDiCE-Culture's nine cultural categories (public holidays, food, geography, history, literature, religion, traditions, clothing, public figures) [Q15, Q122] are structurally aligned with the deployment context and constitute a clear strength. Most cognitive tasks use closed multiple-choice with 2–5 candidates [Q113, Q115], which limits construct richness but is transparent. Two concerns: (i) the dialectal generation task was collapsed from free-form to selection [Q57], which fundamentally reshapes the output space and reduces external validity for an assistant expected to produce dialectal responses; (ii) TruthfulQA's category structure was carried over unmodified [Q121] with near-random performance on dialectal versions [Q73], suggesting the taxonomy may not carve joints that are meaningful in the EGY/LEV context. The qualitative finding that Mistral produced a culturally non-compliant answer (mixed-gender dancing in Qatar) [Q167] illustrates how Western-derived generation criteria can misfire; however, this concerns Gulf rather than EGY/LEV specifically. BoolQ's yes/no structure is preserved by explicit instruction [Q205, Q217].

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q15  | "Our dataset includes questions on public holidays, food, geography, history, public figures, traditional clothing, and more." (p.2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Q57  | "To address this, we simplified the task: models were given multiple response options, with only one in the correct dialect, and asked to select the best response." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Q73  | "The results for the TruthfulQA task across Lev, Egy, and MSA show that model performances are often close to random, especially for Lev and Egy, as reported in Figure 7." (p.7)                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Q113 | "Each question in the dataset has 2-5 candidate answers, with one correct option." (p.16)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Q115 | "The underlying task for this dataset is multiple-choice question answering—given a question q and two possible solutions s1 and s2, the task is to choose the most appropriate solution, of which exactly one is correct." (p.16)                                                                                                                                                                                                                                                                                                                                                                     |
| Q121 | "For assessing misinformation, we used TruthfulQA (Lin et al., 2022). This dataset is specifically designed to evaluate the accuracy and truthfulness of answers provided by language models. It includes 817 questions across 28 diverse categories, such as health, law, finance, and politics. The questions are crafted to address the potential for incorrect answers stemming from common misconceptions or false beliefs." (p.17)                                                                                                                                                               |
| Q122 | "In this paper, we introduce a novel benchmark focused on cultural awareness across the Levant, Egypt, and Gulf regions. This benchmark goes beyond language proficiency to assess whether LLMs can accurately capture the cultural nuances specific to these regions. The dataset contains 180 questions covering 9 cultural aspects, including events (e.g., public holidays), traditions (e.g., food and clothing), geography, history, literature, and religion. To our knowledge, this is the first cultural dataset specifically designed to capture region-wise cultural understanding." (p.17) |
| Q167 | "Mistral's response is particularly worth noting, as not only it generates fictional information, it generated an answer describing a dance where men and women are dancing together, although anyone with some awareness of the country's tradition would know such dance does not comply with the culture of Qatar where segregation between men and women is the norm." (p.21)                                                                                                                                                                                                                      |
| Q205 | "Specific instruction: The type of question should not be changed. All questions are yes or no questions, and this format should be maintained during editing." (p.29)                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Q217 | "For example, for editing BoolQ samples, it is important that the question remains a yes/no question." (p.31)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Strengths

- AraDiCE-Culture's 9-category cultural taxonomy is region-grounded and meaningful for EGY/LEV users [Q15, Q122]

- Explicit preservation of dataset-specific output structure (BoolQ yes/no) in post-editing [Q205, Q217]
- Qualitative evaluation of cultural output appropriateness in generation setup surfaces validity issues empirically [Q164, Q165, Q166, Q167]

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q15  | "Our dataset includes questions on public holidays, food, geography, history, public figures, traditional clothing, and more." (p.2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Q122 | "In this paper, we introduce a novel benchmark focused on cultural awareness across the Levant, Egypt, and Gulf regions. This benchmark goes beyond language proficiency to assess whether LLMs can accurately capture the cultural nuances specific to these regions. The dataset contains 180 questions covering 9 cultural aspects, including events (e.g., public holidays), traditions (e.g., food and clothing), geography, history, literature, and religion. To our knowledge, this is the first cultural dataset specifically designed to capture region-wise cultural understanding." (p.17) |
| Q164 | "With a simple geography question related to one of the Levant countries (Q1), we observe that Jais and AceGPT respond with the correct answer, while Llama 3 partially answer the question providing additional irrelevant and incorrect information." (p.21)                                                                                                                                                                                                                                                                                                                                         |
| Q165 | "Mistral shows a different behavior, hallucinating and responding with non-existent geographical locations." (p.21)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Q166 | "Similar trends can be observed with Q2 about one of the traditions of a Gulf country, where Jais and AceGPT both correctly answer the question, and Llama 3 returns an answer that is more of a description or explanation." (p.21)                                                                                                                                                                                                                                                                                                                                                                   |
| Q167 | "Mistral's response is particularly worth noting, as not only it generates fictional information, it generated an answer describing a dance where men and women are dancing together, although anyone with some awareness of the country's tradition would know such dance does not comply with the culture of Qatar where segregation between men and women is the norm." (p.21)                                                                                                                                                                                                                      |
| Q205 | "Specific instruction: The type of question should not be changed. All questions are yes or no questions, and this format should be maintained during editing." (p.29)                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Q217 | "For example, for editing BoolQ samples, it is important that the question remains a yes/no question." (p.31)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                          |
|------|--------------------------------------------------------------|
| OO-1 | Review output label categories for regional relevance        |
| OO-2 | Identify missing regionally specific categories              |
| OO-3 | Flag categories encoding non-regional values                 |
| OO-4 | Consider stakeholder-driven taxonomy redesign                |
| OO-5 | Document taxonomy issues harming structural/content validity |

### Checklist responses:

**OO-1:** Cultural benchmark's 9 categories align with regional relevance [Q15, Q122]. MCQ structures are task-appropriate but constrained [Q113]. TruthfulQA categories were not re-evaluated for regional relevance [Q121].

**OO-2:** Missing categories: free-form dialectal generation as a first-class output taxonomy [Q57]; culturally appropriate refusal / sensitivity handling is not a labeled output category (INSUFFICIENT DOCUMENTATION); sub-Levantine variety labels for dialect ID (INSUFFICIENT DOCUMENTATION, see WEB-17).

**OO-3:** TruthfulQA categories reflect Western misinformation taxonomies not adapted for regional context [Q121, Q73]. Cultural evaluation via generation revealed Western-taxonomy failures (Mistral's mixed-gender dancing example [Q167]) though this is Gulf-specific.

**OO-4:** AraDiCE-Culture represents a partial stakeholder-driven redesign via native annotators [Q24, Q25]. No taxonomy redesign was performed for TruthfulQA or other translated cognitive tasks.

**OO-5:** Taxonomy issues harming validity: (a) collapse of generation output space to selection [Q57] harms structural validity for assistant deployment; (b) unmodified TruthfulQA category structure [Q121] risks content-validity mismatch; (c) monolithic 'Levantine' treatment obscures sub-variety structure [WEB-17].

| ID | Evidence |
|----|----------|
|----|----------|

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q15    | "Our dataset includes questions on public holidays, food, geography, history, public figures, traditional clothing, and more." (p.2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Q24    | "Cultural Dataset: We curated a dataset of 180 culturally specific questions by hiring native annotators from the Gulf, Egypt, and the Levant regions to generate seed questions centered on cultural and country-specific themes." (p.3)                                                                                                                                                                                                                                                                                                                                                              |
| Q25    | "After reviewing all submitted questions, we selected those with varying answers across regions. To verify these differences, we appended a country name to each question and used Google Search to examine the top 5 search results. Only questions with distinct answers across countries were retained, resulting in a final set of 30 unique questions." (p.3)                                                                                                                                                                                                                                     |
| Q57    | "To address this, we simplified the task: models were given multiple response options, with only one in the correct dialect, and asked to select the best response." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Q73    | "The results for the TruthfulQA task across Lev, Egy, and MSA show that model performances are often close to random, especially for Lev and Egy, as reported in Figure 7." (p.7)                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Q113   | "Each question in the dataset has 2-5 candidate answers, with one correct option." (p.16)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Q121   | "For assessing misinformation, we used TruthfulQA (Lin et al., 2022). This dataset is specifically designed to evaluate the accuracy and truthfulness of answers provided by language models. It includes 817 questions across 28 diverse categories, such as health, law, finance, and politics. The questions are crafted to address the potential for incorrect answers stemming from common misconceptions or false beliefs." (p.17)                                                                                                                                                               |
| Q122   | "In this paper, we introduce a novel benchmark focused on cultural awareness across the Levant, Egypt, and Gulf regions. This benchmark goes beyond language proficiency to assess whether LLMs can accurately capture the cultural nuances specific to these regions. The dataset contains 180 questions covering 9 cultural aspects, including events (e.g., public holidays), traditions (e.g., food and clothing), geography, history, literature, and religion. To our knowledge, this is the first cultural dataset specifically designed to capture region-wise cultural understanding." (p.17) |
| Q167   | "Mistral's response is particularly worth noting, as not only it generates fictional information, it generated an answer describing a dance where men and women are dancing together, although anyone with some awareness of the country's tradition would know such dance does not comply with the culture of Qatar where segregation between men and women is the norm." (p.21)                                                                                                                                                                                                                      |
| WEB-17 | DialectalArabicMMLU separates Syrian dialect from other Levantine, evidencing that a finer output taxonomy for Levantine sub-varieties is possible and reveals gaps AraDiCE's monolithic Levantine label conceals ( <a href="https://arxiv.org/abs/2510.27543">https://arxiv.org/abs/2510.27543</a> )                                                                                                                                                                                                                                                                                                  |

## Information Gaps

- Whether culturally appropriate refusal or safety-linked output categories are represented
- Whether AraDiCE-Culture's 9 categories exhaustively cover assistant-relevant cultural query types

## Requires Expert Verification

- Whether TruthfulQA's carried-over category structure is coherent for EGY/LEV misinformation patterns

## Remediation

**Gap 1: Monolithic 'Levantine' label obscures Syrian/Lebanese/Jordanian/Palestinian sub-variety gaps [WEB-17]; TruthfulQA categories not adapted regionally [Q121]**

**Recommendation:** Adopt Levantine sub-variety labels (following DialectalArabicMMLU [WEB-17]); redesign or supplement TruthfulQA categories with regionally-salient misinformation themes (health, political, conflict-related).

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q121   | "For assessing misinformation, we used TruthfulQA (Lin et al., 2022). This dataset is specifically designed to evaluate the accuracy and truthfulness of answers provided by language models. It includes 817 questions across 28 diverse categories, such as health, law, finance, and politics. The questions are crafted to address the potential for incorrect answers stemming from common misconceptions or false beliefs." (p.17) |
| WEB-17 | DialectalArabicMMLU separates Syrian dialect from other Levantine, evidencing that a finer output taxonomy for Levantine sub-varieties is possible and reveals gaps AraDiCE's monolithic Levantine label conceals ( <a href="https://arxiv.org/abs/2510.27543">https://arxiv.org/abs/2510.27543</a> )                                                                                                                                    |

# Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: ARADICE | Context: Egyptian and Levantine Arabic General-Purpose Assistant Users

**Output Content definition:** Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The post-editing workforce is well-documented and substantial for the translated benchmarks: 32 native EGY/LEV/MSA speakers, many professional translators, dataset-specific guidelines and training, third-party competitive compensation, and expert-translator random-sample review across ~45K post-edited items [Q34, Q35, Q37, Q38, Q39]. AraDiCE-Culture followed the NativeQA framework combining web search with cultural knowledge for gold answers [Q27, Q206]. However, several serious issues undermine label validity for the target region: (a) no inter-annotator agreement statistics are reported for any dataset (documented gap); (b) the cultural dataset's MSA/dialect annotation was done primarily by the paper authors themselves with no compensation, external annotators only for Egyptian [Q212, Q213, Q214] — a plausible bias source explicitly flagged in the region context; (c) quality-assurance review was not applied uniformly due to resource constraints [Q232]; (d) Egyptian required significantly more post-editing effort than Levantine [Q230], suggesting uneven label quality between the two target dialects; (e) 'Finding such native speakers was also a challenge' [Q227]. The one-translator-per-item assignment [Q36] combined with absent IAA means label reliability is not empirically demonstrated.

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                        |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q27  | "For gold-standard answers, we followed the NativeQA framework (Hasan et al., 2024). Annotators reviewed both the questions and the top 5 search results, combining web data with their cultural knowledge to provide the most accurate answers." (p.3)                                                                                                                                                         |
| Q34  | "Team and Setup The translation team comprised 32 native speakers fluent in Levantine, Egyptian, and MSA, with educational backgrounds ranging from Bachelor's to Master's degrees, and ages between 21 and 53." (p.4)                                                                                                                                                                                          |
| Q35  | "Many were professional translators. They received specific guidelines and training tailored to each dataset and dialect to ensure translation quality." (p.4)                                                                                                                                                                                                                                                  |
| Q36  | "To manage the post-editing workload efficiently, we assigned one translator per item, which helped minimize costs and time." (p.4)                                                                                                                                                                                                                                                                             |
| Q37  | "A random sample of post-edited texts was reviewed by the expert translators for quality assurance." (p.4)                                                                                                                                                                                                                                                                                                      |
| Q38  | "The process was streamlined using an in-house annotation platform and 17 dedicated projects, resulting in the post-editing of ~45K items." (p.4)                                                                                                                                                                                                                                                               |
| Q39  | "A third-party company handled the hiring and competitive compensation of translators." (p.4)                                                                                                                                                                                                                                                                                                                   |
| Q206 | "For this annotation task, the following two information is provided to annotators. 1. A question written in a dialect (e.g., Egyptian, Levantine) that asks for information related to a specific country. 2. A list of a maximum 5 URLs of web pages that potentially have an answer to that question. Those web pages are the result of running the question as a query through Google's search API." (p.30) |
| Q212 | "For this annotation task, we did not use any annotation platform. Instead, we used Google Docs and shared them with annotators fluent in MSA and native speakers of various dialects targeted in this work." (p.31)                                                                                                                                                                                            |
| Q213 | "The annotators for this task are primarily the authors of the papers; therefore, no compensation was provided." (p.31)                                                                                                                                                                                                                                                                                         |
| Q214 | "Only for the Egyptian part of the dataset, external annotators participated." (p.31)                                                                                                                                                                                                                                                                                                                           |
| Q227 | "For post editing process, we hired native speakers of specific dialects. This is important to ensure a good quality post-editing. Finding such native speakers was also a challenge for us." (p.32)                                                                                                                                                                                                            |
| Q230 | "When comparing post-editing for dialects, we observed that Egyptian Arabic required significantly more edits, which in turn increased the time and effort needed." (p.32)                                                                                                                                                                                                                                      |
| Q232 | "However, due to limited resources, we could not maintain this level of review across all datasets." (p.33)                                                                                                                                                                                                                                                                                                     |

## Strengths

- 32 native EGY/LEV/MSA post-editors, many professional translators, third-party competitive compensation [Q34, Q35, Q39]
- Detailed per-dataset guidelines and training [Q32, Q33]
- Expert-translator QA review of random samples across ~45K items [Q37, Q38]
- AraDiCE-Culture gold answers grounded in NativeQA framework with search + native cultural knowledge [Q27, Q206]
- Post-editing guidelines explicitly address Arabic-specific correctness (Hamzah, gender/number agreement, culturally sensitive Winogrande items) [Q171, Q172, Q173, Q198, Q223]

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                        |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q27  | "For gold-standard answers, we followed the NativeQA framework (Hasan et al., 2024). Annotators reviewed both the questions and the top 5 search results, combining web data with their cultural knowledge to provide the most accurate answers." (p.3)                                                                                                                                                         |
| Q32  | "Guidelines To assist the translators and maintain dataset integrity, we provided two sets of guidelines for each dataset (except Arabic MMLU). One set focused on post-editing translated samples in MSA, while the other targeted the same process for dialects." (p.4)                                                                                                                                       |
| Q33  | "Each dataset's guidelines included: (i) details on the task components (e.g., one question and four answers), (ii) general instructions for correcting errors and improving fluency and adequacy, and (iii) specific instructions for handling unique cases within the datasets." (p.4)                                                                                                                        |
| Q34  | "Team and Setup The translation team comprised 32 native speakers fluent in Levantine, Egyptian, and MSA, with educational backgrounds ranging from Bachelor's to Master's degrees, and ages between 21 and 53." (p.4)                                                                                                                                                                                          |
| Q35  | "Many were professional translators. They received specific guidelines and training tailored to each dataset and dialect to ensure translation quality." (p.4)                                                                                                                                                                                                                                                  |
| Q37  | "A random sample of post-edited texts was reviewed by the expert translators for quality assurance." (p.4)                                                                                                                                                                                                                                                                                                      |
| Q38  | "The process was streamlined using an in-house annotation platform and 17 dedicated projects, resulting in the post-editing of ≈45K items." (p.4)                                                                                                                                                                                                                                                               |
| Q39  | "A third-party company handled the hiring and competitive compensation of translators." (p.4)                                                                                                                                                                                                                                                                                                                   |
| Q171 | "Instruction 1: When necessary, the text should be rewritten to adhere to the syntactic and semantic structure of Arabic." (p.22)                                                                                                                                                                                                                                                                               |
| Q172 | "For MSA, it is also important to correct the wrong use of Hamzah (Hamzatul-wasl and Hamzatul-Qata'a)." (p.22)                                                                                                                                                                                                                                                                                                  |
| Q173 | "Fixing the use of inappropriate verbs that do not align between sentences and the options associated with them in terms of their grammatical gender and singularity or plurality." (p.22)                                                                                                                                                                                                                      |
| Q198 | "Specific instruction: If necessary, the options should be edited to ensure alignment with the sentence in terms of gender, singularity, or plurality. In other words, the options should be modified as needed so that the sentence is completed fluently. Grammatical or semantic adjustments may be required to achieve this." (p.29)                                                                        |
| Q206 | "For this annotation task, the following two information is provided to annotators. 1. A question written in a dialect (e.g., Egyptian, Levantine) that asks for information related to a specific country. 2. A list of a maximum 5 URLs of web pages that potentially have an answer to that question. Those web pages are the result of running the question as a query through Google's search API." (p.30) |
| Q223 | "During our analysis of the datasets, we noticed that some samples in Winogrande conflicted with the religion and culture of the Arab region. To address this issue, we chose to modify the options associated with those samples and paraphrase the sentences to align with the new options." (p.32)                                                                                                           |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                               |
|------|-------------------------------------------------------------------|
| OC-1 | Do ground-truth labels reflect regional stakeholder perspectives? |
| OC-2 | Assess potential annotator-regional population disagreement       |
| OC-3 | Review annotator demographics from documentation                  |
| OC-4 | Consider label re-annotation by regional annotators               |
| OC-5 | Review aggregation methods for minority perspective erasure       |

|      |                                                            |
|------|------------------------------------------------------------|
| OC-6 | Document label issues harming convergent/external validity |
|------|------------------------------------------------------------|

## Checklist responses:

**OC-1:** Partially. For AraDiCE-Culture, gold answers reflect regional stakeholder perspectives via native annotators + NativeQA [Q24, Q27]. For translated cognitive benchmarks, ground truth is inherited from English sources and post-edited by native speakers [Q34]; underlying label semantics remain Western-derived [Q99, Q224].

**OC-2:** Likely disagreement risk in TruthfulQA (Western misinformation targets [Q121, Q73]) and in Winogrande items with religious/cultural conflict that were only partially edited [Q223]. Cultural annotation done by authors for non-Egyptian parts is a documented bias risk [Q212, Q213, Q214].

**OC-3:** Demographics documented: 32 native EGY/LEV/MSA speakers, Bachelor's–Master's, ages 21–53, many professional translators [Q34, Q35]. Third-party hiring [Q39]. No further demographic breakdown (e.g., country of origin, gender, sub-variety) reported.

**OC-4:** One translator per item [Q36]; no re-annotation or IAA reported for any dataset (documented gap). Random QA review by expert translators applied non-uniformly [Q37, Q232].

**OC-5:** Aggregation methods not detailed — single-translator assignment means no aggregation across annotators for individual items [Q36]. This structurally erases minority-perspective signal.

**OC-6:** Label issues harming validity: (a) no IAA statistics for any dataset; (b) authors-as-annotators for cultural dataset [Q212, Q213, Q214]; (c) non-uniform QA review [Q232]; (d) more edits required for Egyptian [Q230] suggesting uneven Egyptian label quality; (e) single-annotator assignment [Q36].

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q24  | "Cultural Dataset: We curated a dataset of 180 culturally specific questions by hiring native annotators from the Gulf, Egypt, and the Levant regions to generate seed questions centered on cultural and country-specific themes." (p.3)                                                                                                                                                                                                |
| Q27  | "For gold-standard answers, we followed the NativeQA framework (Hasan et al., 2024). Annotators reviewed both the questions and the top 5 search results, combining web data with their cultural knowledge to provide the most accurate answers." (p.3)                                                                                                                                                                                  |
| Q34  | "Team and Setup The translation team comprised 32 native speakers fluent in Levantine, Egyptian, and MSA, with educational backgrounds ranging from Bachelor's to Master's degrees, and ages between 21 and 53." (p.4)                                                                                                                                                                                                                   |
| Q35  | "Many were professional translators. They received specific guidelines and training tailored to each dataset and dialect to ensure translation quality." (p.4)                                                                                                                                                                                                                                                                           |
| Q36  | "To manage the post-editing workload efficiently, we assigned one translator per item, which helped minimize costs and time." (p.4)                                                                                                                                                                                                                                                                                                      |
| Q37  | "A random sample of post-edited texts was reviewed by the expert translators for quality assurance." (p.4)                                                                                                                                                                                                                                                                                                                               |
| Q39  | "A third-party company handled the hiring and competitive compensation of translators." (p.4)                                                                                                                                                                                                                                                                                                                                            |
| Q73  | "The results for the TruthfulQA task across Lev, Egy, and MSA show that model performances are often close to random, especially for Lev and Egy, as reported in Figure 7." (p.7)                                                                                                                                                                                                                                                        |
| Q99  | "Another issue to highlight is that most datasets, except ArabicMMLU, are adapted from English and thus are influenced by Western culture. While we made efforts in annotation guidelines and post-editing to address these cultural biases, the subjective nature of sensitivity means that some samples may still be considered sensitive by different individuals or communities." (p.10)                                             |
| Q121 | "For assessing misinformation, we used TruthfulQA (Lin et al., 2022). This dataset is specifically designed to evaluate the accuracy and truthfulness of answers provided by language models. It includes 817 questions across 28 diverse categories, such as health, law, finance, and politics. The questions are crafted to address the potential for incorrect answers stemming from common misconceptions or false beliefs." (p.17) |
| Q212 | "For this annotation task, we did not use any annotation platform. Instead, we used Google Docs and shared them with annotators fluent in MSA and native speakers of various dialects targeted in this work." (p.31)                                                                                                                                                                                                                     |
| Q213 | "The annotators for this task are primarily the authors of the papers; therefore, no compensation was provided." (p.31)                                                                                                                                                                                                                                                                                                                  |
| Q214 | "Only for the Egyptian part of the dataset, external annotators participated." (p.31)                                                                                                                                                                                                                                                                                                                                                    |
| Q223 | "During our analysis of the datasets, we noticed that some samples in Winogrande conflicted with the religion and culture of the Arab region. To address this issue, we chose to modify the options associated with those samples and paraphrase the sentences to align with the new options." (p.32)                                                                                                                                    |
| Q224 | "In all the datasets that were post-edited, except for ArabicMMLU, we consistently observed that the samples were designed with a focus on Western culture." (p.32)                                                                                                                                                                                                                                                                      |

|      |                                                                                                                                                                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q230 | "When comparing post-editing for dialects, we observed that Egyptian Arabic required significantly more edits, which in turn increased the time and effort needed." (p.32) |
| Q232 | "However, due to limited resources, we could not maintain this level of review across all datasets." (p.33)                                                                |

## Information Gaps

- No inter-annotator agreement statistics reported for any dataset
- Annotator demographic breakdown by country of origin, gender, Levantine sub-variety not reported
- Coverage rate of the random QA review is not quantified

## Requires Expert Verification

- Whether authors-as-annotators for non-Egyptian cultural items introduces systematic bias in gold answers
- Whether label quality parity between EGY and LEV is acceptable for comparative evaluation given Egyptian required more edits

## Remediation

**Gap 1: No inter-annotator agreement statistics; single translator per item; authors-as-annotators for parts of cultural dataset; uneven EGY vs. LEV post-editing effort [Q36, Q212–Q214, Q230, Q232]**

**Recommendation:** Compute IAA on a stratified sample of items using independent EGY/LEV native raters; re-annotate the author-annotated cultural items with compensated external annotators; report label-quality parity between EGY and LEV explicitly.

| ID   | Evidence                                                                                                                                                                                                             |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q36  | "To manage the post-editing workload efficiently, we assigned one translator per item, which helped minimize costs and time." (p.4)                                                                                  |
| Q212 | "For this annotation task, we did not use any annotation platform. Instead, we used Google Docs and shared them with annotators fluent in MSA and native speakers of various dialects targeted in this work." (p.31) |
| Q213 | "The annotators for this task are primarily the authors of the papers; therefore, no compensation was provided." (p.31)                                                                                              |
| Q214 | "Only for the Egyptian part of the dataset, external annotators participated." (p.31)                                                                                                                                |
| Q230 | "When comparing post-editing for dialects, we observed that Egyptian Arabic required significantly more edits, which in turn increased the time and effort needed." (p.32)                                           |
| Q232 | "However, due to limited resources, we could not maintain this level of review across all datasets." (p.33)                                                                                                          |

# Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: ARADICE | Context: Egyptian and Levantine Arabic General-Purpose Assistant Users

**Output Form definition:** Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Evaluation uses appropriate per-task metrics — F1 for dialect ID, normalized accuracy for cognitive MCQ, SacreBLEU for MT [Q50] — through the LM Evaluation Harness [Q49], with human evaluation supplementing BLEU for MT system selection [Q141, Q142, Q143]. Qualitative error analyses on dialect ID [Q53] and cultural generation [Q151, Q164–Q167] add validity signal beyond aggregate metrics. However, four significant limitations reduce external validity for the assistant deployment: (i) zero-shot only [Q46, Q47], while production assistants operate in instruction-tuned/few-shot regimes; (ii) models capped at ≤13B parameters [Q101], excluding the larger models typically used in production; (iii) extended evaluation with newer models covered only ArabicMMLU and PIQA due to resource constraints [Q83]; (iv) BLEU is acknowledged as a risk for overfitting [Q139] and, per the region context, LLM-as-judge approaches for Arabic dialectal generation carry systematic pro-MSA bias [WEB-16]. Output modality is text-only, matching text-assistant deployment but not speech.

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q46  | "In our experiments, we employed zero-shot learning to observe performance differences across dialects." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Q47  | "Although few-shot learning is known to enhance performance, we chose not to include it to simplify the experiments and minimize computational costs." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Q49  | "We used the LM Evaluation Harness for both generation and multiple-choice tasks, employing standard metrics for each task and dataset." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Q50  | "We used F1 scores for dialect identification, normalized accuracy for cognitive tasks, and SacreBLEU (Post, 2018) for machine translation." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Q53  | "Our error analysis (Figure 4) shows Llama 3 confuse Gulf with Lev, Mistral confuses MSA with Lev/Gulf, Jais often confuse Egy with Gulf, while AceGPT mistakes MSA for Gulf." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Q83  | "Due to resource constraints, we focus on Arabic MMLU and PIQA datasets, which assess performance on diverse, challenging tasks." (p.8)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Q101 | "Due to resource limitations, we only evaluated our benchmarks using models up to 13B parameters. As a one-off experiment, we tested the ArabicMMLU task with a larger Jais 30B, model. We found (See results in Appendix D.1, Figure 15) Jais 30B to perform similarly to Jais 13B indicating that larger models do not necessarily show significant improvements in this case. Due to hardware limitations, we could not run Jais 70B models, but it would be interesting to compare the higher-scale Jais and Llama models to see if their increased scale can compensate for the lack of dialect-specific training." (p.10) |
| Q139 | "Nonetheless, basing decisions solely on BLEU scores presents a risk of over-fitting." (p.18)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Q141 | "Therefore, we carried out a human evaluation of the selected systems to select the best model for translation." (p.18)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Q142 | "For this, we selected a sample of 100 sentences from different genres, and carried out translations using the systems." (p.18)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Q143 | "We conducted a human evaluation by showing the output of three translation systems and asked the participants to select the "best system(s)"." (p.18)                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Q151 | "We conducted qualitative analysis on the generation task of the models by prompting them in the dialectal version of "How would you respond to this phrase in the X dialect", where X could be one of the Egyptian, Levantine, or Gulf dialects (examples on models' response can be found in Table 12)." (p.19)                                                                                                                                                                                                                                                                                                               |
| Q164 | "With a simple geography question related to one of the Levant countries (Q1), we observe that Jais and AceGPT respond with the correct answer, while Llama 3 partially answer the question providing additional irrelevant and incorrect information." (p.21)                                                                                                                                                                                                                                                                                                                                                                  |
| Q165 | "Mistral shows a different behavior, hallucinating and responding with non-existent geographical locations." (p.21)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|               |                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q166</b>   | "Similar trends can be observed with Q2 about one of the traditions of a Gulf country, where Jais and AceGPT both correctly answer the question, and Llama 3 returns an answer that is more of a description or explanation." (p.21)                                                                                                                                              |
| <b>Q167</b>   | "Mistral's response is particularly worth noting, as not only it generates fictional information, it generated an answer describing a dance where men and women are dancing together, although anyone with some awareness of the country's tradition would know such dance does not comply with the culture of Qatar where segregation between men and women is the norm." (p.21) |
| <b>WEB-16</b> | GPT-4 as evaluator systematically scores MSA higher than dialect, cautioning against naive LLM-as-judge use ( <a href="https://arxiv.org/pdf/2407.07551">https://arxiv.org/pdf/2407.07551</a> )                                                                                                                                                                                   |

## Strengths

- Task-appropriate metric selection (F1, normalized accuracy, SacreBLEU) via LM Eval Harness [Q49, Q50]
- Human evaluation supplements BLEU for MT system selection [Q141, Q142, Q143]
- Qualitative analyses on dialect ID and cultural generation provide richer validity evidence than aggregate metrics alone [Q53, Q151, Q164–Q167]
- Prompts and configurations released to support reproducibility [Q48, Q96, Q97]

| ID          | Evidence                                                                                                                                                                                                                                                                                                                                                                          |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q48</b>  | "We used prompts in English, MSA, and dialects depending on the task, and released both the prompts and configurations via lm-harness." (p.5)                                                                                                                                                                                                                                     |
| <b>Q49</b>  | "We used the LM Evaluation Harness for both generation and multiple-choice tasks, employing standard metrics for each task and dataset." (p.5)                                                                                                                                                                                                                                    |
| <b>Q50</b>  | "We used F1 scores for dialect identification, normalized accuracy for cognitive tasks, and SacreBLEU (Post, 2018) for machine translation." (p.5)                                                                                                                                                                                                                                |
| <b>Q53</b>  | "Our error analysis (Figure 4) shows Llama 3 confuse Gulf with Lev, Mistral confuses MSA with Lev/Gulf, Jais often confuse Egy with Gulf, while AceGPT mistakes MSA for Gulf." (p.5)                                                                                                                                                                                              |
| <b>Q96</b>  | "We have released our dialectal benchmarks and models to support future research and advancements in NLP for low-resource languages." (p.9)                                                                                                                                                                                                                                       |
| <b>Q97</b>  | "We also released the dialectal translation models and benchmarks developed in this study to support further research in NLP for low-resource languages." (p.9)                                                                                                                                                                                                                   |
| <b>Q141</b> | "Therefore, we carried out a human evaluation of the selected systems to select the best model for translation." (p.18)                                                                                                                                                                                                                                                           |
| <b>Q142</b> | "For this, we selected a sample of 100 sentences from different genres, and carried out translations using the systems." (p.18)                                                                                                                                                                                                                                                   |
| <b>Q143</b> | "We conducted a human evaluation by showing the output of three translation systems and asked the participants to select the "best system(s)." (p.18)                                                                                                                                                                                                                             |
| <b>Q151</b> | "We conducted qualitative analysis on the generation task of the models by prompting them in the dialectal version of "How would you respond to this phrase in the X dialect", where X could be one of the Egyptian, Levantine, or Gulf dialects (examples on models' response can be found in Table 12)." (p.19)                                                                 |
| <b>Q164</b> | "With a simple geography question related to one of the Levant countries (Q1), we observe that Jais and AceGPT respond with the correct answer, while Llama 3 partially answer the question providing additional irrelevant and incorrect information." (p.21)                                                                                                                    |
| <b>Q165</b> | "Mistral shows a different behavior, hallucinating and responding with non-existent geographical locations." (p.21)                                                                                                                                                                                                                                                               |
| <b>Q166</b> | "Similar trends can be observed with Q2 about one of the traditions of a Gulf country, where Jais and AceGPT both correctly answer the question, and Llama 3 returns an answer that is more of a description or explanation." (p.21)                                                                                                                                              |
| <b>Q167</b> | "Mistral's response is particularly worth noting, as not only it generates fictional information, it generated an answer describing a dance where men and women are dancing together, although anyone with some awareness of the country's tradition would know such dance does not comply with the culture of Qatar where segregation between men and women is the norm." (p.21) |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                   |
|-------------|-------------------------------------------------------|
| <b>OF-1</b> | Does output modality match regional deployment needs? |
| <b>OF-2</b> | Assess text-to-speech availability for speech output  |

|      |                                                        |
|------|--------------------------------------------------------|
| OF-3 | Consider literacy rates and accessibility requirements |
| OF-4 | Document form mismatches harming external validity     |

### Checklist responses:

**OF-1:** Output form is text, matching a text-based assistant. It does not match voice-enabled deployments (INSUFFICIENT DOCUMENTATION on any speech output evaluation).

**OF-2:** TTS availability not addressed by the benchmark; NADI 2025 exists as external complement [WEB-15].

**OF-3:** Text-only outputs are broadly compatible with the target population's literacy profile (Lebanon 92% [WEB-11]; Egypt ~73–76% pending verification [WEB-10]; Jordan reported ~98% pending verification [WEB-12]). However, lower literacy populations, particularly in rural Upper Egypt or displaced Syrian populations, may be underserved by text-only evaluation.

**OF-4:** Form-related validity limitations: (a) zero-shot only [Q46, Q47]; (b)  $\leq 13\text{B}$  parameter model cap [Q101]; (c) extended evaluation only on 2 tasks [Q83]; (d) BLEU overfitting risk [Q139]; (e) no evaluation of speech output form.

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q46    | "In our experiments, we employed zero-shot learning to observe performance differences across dialects." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Q47    | "Although few-shot learning is known to enhance performance, we chose not to include it to simplify the experiments and minimize computational costs." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Q49    | "We used the LM Evaluation Harness for both generation and multiple-choice tasks, employing standard metrics for each task and dataset." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Q50    | "We used F1 scores for dialect identification, normalized accuracy for cognitive tasks, and SacreBLEU (Post, 2018) for machine translation." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Q83    | "Due to resource constraints, we focus on Arabic MMLU and PIQA datasets, which assess performance on diverse, challenging tasks." (p.8)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Q101   | "Due to resource limitations, we only evaluated our benchmarks using models up to 13B parameters. As a one-off experiment, we tested the ArabicMMLU task with a larger Jais 30B, model. We found (See results in Appendix D.1, Figure 15) Jais 30B to perform similarly to Jais 13B indicating that larger models do not necessarily show significant improvements in this case. Due to hardware limitations, we could not run Jais 70B models, but it would be interesting to compare the higher-scale Jais and Llama models to see if their increased scale can compensate for the lack of dialect-specific training." (p.10) |
| Q139   | "Nonetheless, basing decisions solely on BLEU scores presents a risk of over-fitting." (p.18)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| WEB-10 | Egypt literacy indicator (approx 73–76%, pending verification) — text-only outputs may be less accessible for lower-literacy sub-populations ( <a href="https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=EG">https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=EG</a> )                                                                                                                                                                                                                                                                                                                                |
| WEB-11 | Lebanon literacy 92% — supports text-output form compatibility for that sub-population ( <a href="https://www.theglobaleconomy.com/Lebanon/Literacy_rate/">https://www.theglobaleconomy.com/Lebanon/Literacy_rate/</a> )                                                                                                                                                                                                                                                                                                                                                                                                        |
| WEB-12 | <a href="https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=JO">https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=JO</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| WEB-15 | NADI 2025 addresses spoken output form gap for voice-enabled deployment ( <a href="https://arxiv.org/html/2509.02038">https://arxiv.org/html/2509.02038</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| WEB-16 | GPT-4 as evaluator systematically scores MSA higher than dialect, cautioning against naive LLM-as-judge use ( <a href="https://arxiv.org/pdf/2407.07551">https://arxiv.org/pdf/2407.07551</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                 |

### Information Gaps

- Verified Egypt literacy rate and its implications for text-only deployment coverage
- Whether any dialect-aware evaluation of long-form generation form is planned or feasible

### Requires Expert Verification

- Whether zero-shot  $\leq 13\text{B}$  evaluation results transfer to instruction-tuned larger production assistants

### Remediation

**Gap 1: Zero-shot only and  $\leq 13\text{B}$  parameter models limit generalization to production assistants [Q46, Q47, Q101]**

**Recommendation:** Extend evaluation to few-shot / instruction-tuned regimes and larger open + proprietary models (via HELM Arabic [WEB-26]); include human rating alongside BLEU to counter overfitting risk [Q139].

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q46    | "In our experiments, we employed zero-shot learning to observe performance differences across dialects." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Q47    | "Although few-shot learning is known to enhance performance, we chose not to include it to simplify the experiments and minimize computational costs." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Q101   | "Due to resource limitations, we only evaluated our benchmarks using models up to 13B parameters. As a one-off experiment, we tested the ArabicMMLU task with a larger Jais 30B, model. We found (See results in Appendix D.1, Figure 15) Jais 30B to perform similarly to Jais 13B indicating that larger models do not necessarily show significant improvements in this case. Due to hardware limitations, we could not run Jais 70B models, but it would be interesting to compare the higher-scale Jais and Llama models to see if their increased scale can compensate for the lack of dialect-specific training." (p.10) |
| Q139   | "Nonetheless, basing decisions solely on BLEU scores presents a risk of over-fitting." (p.18)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| WEB-26 | Stanford HELM Arabic enables comparison with larger proprietary models beyond AraDiCE's $\leq 13$ B cap ( <a href="https://crfm.stanford.edu/helm/arabic/latest/">https://crfm.stanford.edu/helm/arabic/latest/</a> )                                                                                                                                                                                                                                                                                                                                                                                                           |